

Full-body Avatar Generation for Increased Embodiment

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ABSTRACT

Having a virtual full-body increases the embodiment of users in virtual reality (VR) applications. As consumer-grade virtual reality systems provide the head and two hand-held sensors, full-body motion generation is a vastly under-determined problem. Consequently, most current VR applications show the upper-body only using an inverse kinematics method from three given sensors or wearing the additional sensors on the lower body. Recent studies have focused on neural network methods to generate a full-body avatar from three sparse sensors. Although neural network methods produce notable results with low error metrics in motion capture datasets, it is challenging to apply to online VR systems. Technical issues include that VR sensors fluctuate and lose connection, hand-held controllers orient differently depending on the user's grabbing style, the coordinate system of VR platforms differs from that of motion capture datasets, and the global position and orientation of the users vary in a virtual world. We address these technical issues and present a solution to generate a full-body avatar in VR systems. We compare our method with other full-body generation methods on online VR systems. Then, we show how our method works in two kinds of online VR tasks, a single-user obstacle task and a multi-user catch ball task, and conduct a user study for embodiment and preferences.

Index Terms: Computing methodologies—Computer Graphics—Graphics systems and interfaces—Virtual reality; Computing methodologies—Artificial intelligence—Compute Vision—Motion capture

1 INTRODUCTION

A virtual body is quite compelling for immersive experiences, especially when that body moves like one's own body [6]. Using three sensors of the headset and two hand-held controllers of today's virtual reality (VR) system, it is possible to track and visualize the user's head and hands in a virtual world. Inverse kinematics (IK) methods generate plausible upper-body motions from three sensors and enable gesture interactions with the visualized upper-body of avatars. These virtual upper bodies bring a sense of embodiment [14]. The animation of virtual full bodies further enhances embodiment than virtual upper bodies. Creating a virtual full-body from the three sensors of the VR system is a greatly under-determined problem.

A significant amount of work has addressed generating full-body poses in real-time from sparse trackers in recent years. Previous work has used six sensors with a deep neural network [10, 11, 15, 5, 20, 19, 8], four sensors by attaching an additional sensor on the pelvis

joint [17], and three sensors of consumer-grade devices [2, 4, 3, 7, 18, 16]. Although the existing neural network methods for the full-body pose from three sparse sensors show low error metrics and work fine on motion capture datasets, they generate temporal inconsistency and visually suffer side effects on VR systems. On the other hand, the existing reinforcement learning-based methods present temporally consistent animation of a full body but occasionally fail to track abrupt and non-trained complex motions, and pose inference is not real-time for VR systems [18, 16]. Due to these limitations, the existing methods have difficulties being practically used in VR applications.

In VR, tracked signals of the head-mounted devices (HMDs) and hand-held controllers bounce and spike due to lighting, network, obstacles, hardware, or software problems. The positions and orientations of the hand-held controllers differ from those of the wrist and palm joints and vary depending on the users' grabbing style. Moreover, the global locations and orientations vary across applications and change at run-time. The inference model takes these positions and orientations of the VR sensors as inputs and expects these data are consistent with motion capture datasets. Therefore, it is necessary to filter sensor signals, map three VR sensors and the joints of the body model considering the user's body size and grabbing style, align the coordinates of motion capture and VR systems, and find a common reference frame from three sensors useful for the VR system and motion capture datasets.

We aim for the practical application of neural network methods to generate a full-body avatar from three sparse sensors in the online VR system. We propose VR sensor data processing and a Transformer-based regression network trained on the coordinate-aligned motion capture datasets to generate temporally consistent full-body pose simulation from the head and hand sensors as input, show the effectiveness of our method on online VR systems, and conduct a user study of our method for enhanced embodiment.

2 FULL BODY AVATAR GENERATION

VR systems provide the positions and orientations of the HMDs and hand-held controllers with respect to the world frame. Given a set of sensor positions and orientations in the Cartesian coordinate system, we process the sensor data to remove noisy signals and robustly compute the position and orientations of the head and two wrist joints in the common reference frame for an avatar model. A neural network model generates the full body pose of the avatar from these three joint data.

Specifically, we use the SMPL model [12] to represent and animate our human body pose. We use the first 22 joints defined in the kinematic tree of the SMPL human skeleton and ignore the pose of fingers. We use the unity3D engine for a VR system.

2.1 Sensor Data Processing

Sensor data processing consists of three steps. One is sensor mapping which finds correspondences from the HMD and two hand-held controllers to the head and wrist joints of the body model. Another is to represent the position and orientation of the head and wrist joints with respect to the common reference frame shared with the

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Table 1: Comparison with Existing Methods on AMASS

Test Data	Method	MPJPE(cm)	RootPE(cm)
Transitions,	Ours	5.17	3.89
Human Eva	AvatarPoser	5.1	3.73

learning model. Last is temporal smoothing to stabilize the tracking signal.

The users wear the head-mounted devices and hold two controllers in their hands. Let $\mathbf{s}_{s,t}^w$, 3 by 1 vector, be the position of s sensor at t timeframe in a world frame, and $\mathbf{O}_{s,t}^w$, 3 by 3 matrix, be the orientation of s sensor at t timeframe in a world frame. Given a set of sensor positions and orientations, $\{\mathbf{s}_{s,t}^w, \mathbf{O}_{s,t}^w\}_{s=1:3}$, in the Cartesian coordinate system, we find the head and wrist joints of the body model, $\{\mathbf{p}_{j,t}^w, \mathbf{R}_{j,t}^w\}_{j=\text{head},lw,rw}$, where $\mathbf{p}_{j,t}^w$ and $\mathbf{R}_{j,t}^w$ are the position and orientation of j joint at t timeframe in a world frame. We compute the translational and rotational displacements between these sensors and corresponding joints of our SMPL model at a rest pose and assume these relations stay consistent over time.

2.2 Neural Network Avatar Model

We use the Transformer Encoder network similar to the state-of-the-art AvatarPoser method [?], modify the network to deal with our processed sensor data, and process the training and testing datasets adequate so that the generated pose from the network model is adequate for the game engines such as Unity3D and Unreal. For training and testing, we use an AMASS [13], a large-scale motion capture dataset with diverse poses represented with the SMPL body model [12]. We use all kinds of MoCap datasets in AMASS to train our network model thoroughly. We train our model with PyTorch, adopt the Adam optimizer [9], with a learning rate of 10^{-3} , and decay by a factor of 0.97 per 25 epochs.

3 COMPARISONS WITH OTHER FULL BODY AVATARS

We evaluate our proposed system in generating a plausible pose, given the input from the headset and two hand trackers, and compare against state-of-the-art techniques AvatarPoser [7], six trackers, and Final IK [1]. For the AvatarPoser, we use the available source code provided by the authors on GitHub and then run and test their source code to visualize in the Unity 3D engine. Six trackers require VR users to wear three more trackers on the lower body and generate a full-body avatar using the IK method.

We compare the inference results on Transitions and HumanEva subset of the AMASS. Table 1 reports the numerical results for the considered metrics, MPJPE (Mean Per Joint Position Error), which is the mean Euclidean distance error of all the predicted joints in centimeters with the root joint aligned and RootPE(Root Position Error), which measures the Euclidean distance error of the pelvis joint in centimeters.

We compare our and AvatarPoser’s results on online VR data. Therefore, the avatar’s limbs look bent, and the head of the avatar looks upward or downward. Fig. 1 shows the ground truth(left), ours(middle) and AvatarPoser’s result(right). Since the user is taller than the SMPL model, the height of the headset is higher than the SMPL model, causing the head of the avatar to look up in the right image. The middle image in Fig. 1 is the avatar feeding the network with the mapped data, and the orientation of the right wrist corresponds to that of the ground truth. Without the mapping between the sensors and joints, the orientations of the HMD and the hand-held controllers are considered as those of the head and wrist joints.

4 USER STUDY

The comparisons in Sect. 3 are promising. Our neural networks with VR sensor processing can generate reasonably accurate and

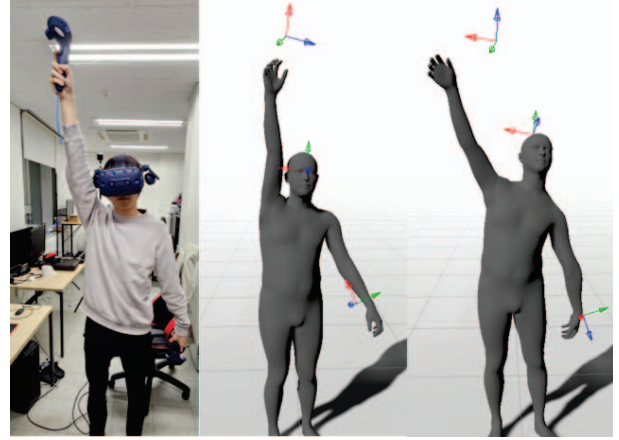


Figure 1: Comparison on online VR dataset; Ground truth(left), ours(middle), and avatarPoser (right). The orientations of the head and wrist joints are drawn in red, green, and blue arrows

plausible poses for a wide set of motions in VR applications. We further conduct a user study to determine how our system and six trackers perform in real consumer-grade VR platforms. The goal of our user study is twofold. First, we want to determine whether our neural network method performs well enough compared to other methods in the task completion perspective. Second, we want to determine whether our neural network method improves the feeling of embodiment.

We used a within-subject design to test our hypotheses and split the study into two tasks. In the first task, the obstacle game, participants must use their bodies to avoid the obstacles. Participants can see their own avatars reflected in the front mirror and receive visual feedback on their avatars. In the second task, the catch ball, two participants need to use their arms and move around to throw and receive the objects to each other. Participants can see each other’s avatars and throw and receive the cube-shaped object to each other. We test two conditions: our neural network and six trackers’ avatars. Ours generates the full-body pose from the HMD and two hand-held controllers as described in Sec. 3. Six trackers’ avatar generates the full-body pose from the HMD, two hand-held controllers, and three additional trackers attached on the pelvis, left foot and right foot joints. Three additional trackers are worn throughout the whole user study to prevent users from guessing different conditions. In both tasks, the order of conditions is randomized. We use the Unity3D game engine, two HTC Vive Pro VR systems, four hand-held controllers, and six additional vice tracker 3.0 trackers. The size of the experimental space is 140cm in width and 120cm in height.

5 RESULTS

We initially recruited 20 participants for the user study. All participants were aged between 22 and 29, 40 % of the participants were female, all participants were right-handed, and 55 % had VR experiences.

In the obstacle task, questions are about a sense of embodiment when participants see their own bodies in the mirror. The questionnaire results are summarized in Fig.2. We observe that the six trackers’ avatar scores higher about a sense of embodiment in all questions from Q1 to Q8. Statistical analysis using a t -test shows no meaningful differences ($p > 0.05$) of the conditions. The neural network and six trackers’ avatars show no significant differences in the sense of embodiment when participants see self-avatars.

In the catch ball task, questions are about a sense of embodiment when participants see another participant’s avatar and throw the balls

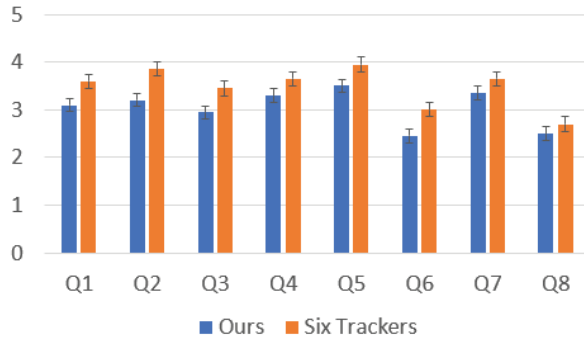


Figure 2: Questionnaire results in the obstacle task. Blue and orange colors depict our and six trackers' methods, respectively

toward the other's avatar. The questionnaire results in Fig.3 and reports that our avatar gets a higher score except for Q3 and Q7. The differences among those questions are insignificant differences after using the t -test with a borderline $p = 0.05$. The neural network and six trackers' avatars show no significant differences in the sense of embodiment when participants see others' avatars in virtual environments.

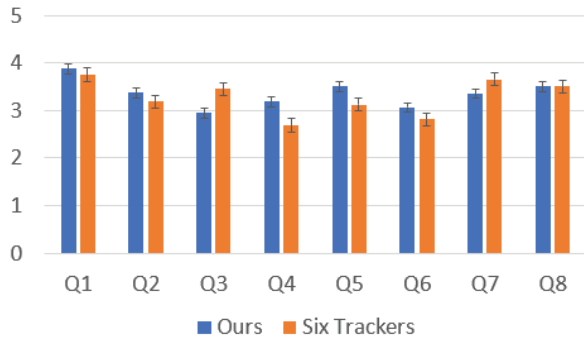


Figure 3: Questionnaire results in the catch ball task. Blue and orange colors depict our and six trackers' methods, respectively

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REFERENCES

- [1] RootMotion Final IK. "https://assetstore.unity.com/packages/tools/animation/final-ik-14290", 2018.
- [2] K. Ahuja, E. Ofek, M. Gonzalez-Franco, C. Holz, and A. D. Wilson. Coolmoves: User motion accentuation in virtual reality. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, 5(2), jun 2021. doi: 10.1145/3463499
- [3] S. Aliakbarian, P. Cameron, F. Bogo, A. Fitzgibbon, and T. J. Cashman. Flag: Flow-based 3d avatar generation from sparse observations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 13253–13262, 2022.
- [4] A. Dittadi, S. Dziadzio, D. Cosker, B. Lundell, T. J. Cashman, and J. Shotton. Full-body motion from a single head-mounted device: generating smpl poses from partial observations. In *Proceedings of the*

- IEEE/CVF International Conference on Computer Vision*, pp. 11687–11697, 2021.
- [5] Y. Huang, M. Kaufmann, E. Aksan, M. J. Black, O. Hilliges, and G. Pons-Moll. Deep inertial poser: Learning to reconstruct human pose from sparse inertial measurements in real time. *ACM Trans. Graph.*, 37(6), dec 2018. doi: 10.1145/3272127.3275108
- [6] J. Jerald. *The VR Book: Human-Centered Design for Virtual Reality*. Association for Computing Machinery and Morgan & Claypool, 2015.
- [7] J. Jiang, P. Strel, H. Qiu, A. Fender, L. Laich, P. Snape, and C. Holz. Avatarposer: Articulated full-body pose tracking from sparse motion sensing. In *Proceedings of European Conference on Computer Vision*. Springer, 2022.
- [8] M. Kaufmann, Y. Zhao, C. Tang, L. Tao, C. Twigg, J. Song, R. Wang, and O. Hilliges. Em-pose: 3d human pose estimation from sparse electromagnetic trackers. In *International Conference on Computer Vision (ICCV)*, 2021.
- [9] D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. In Y. Bengio and Y. LeCun, eds., *3rd International Conference on Learning Representations, ICLR 2015, San Diego, CA, USA, May 7-9, 2015, Conference Track Proceedings*, 2015.
- [10] G. Liu, J. Zhang, W. Wang, and L. McMillan. Human motion estimation from a reduced marker set. In *ACM SIGGRAPH 2006 Sketches, SIGGRAPH '06*, p. 9–es. Association for Computing Machinery, New York, NY, USA, 2006. doi: 10.1145/1179849.1179860
- [11] H. Liu, X. Wei, J. Chai, I. Ha, and T. Rhee. Realtime human motion control with a small number of inertial sensors. In *Symposium on Interactive 3D Graphics and Games, I3D '11*, p. 133–140. Association for Computing Machinery, New York, NY, USA, 2011. doi: 10.1145/1944745.1944768
- [12] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black. Smpl: A skinned multi-person linear model. *ACM Trans. Graph.*, 34(6), nov 2015. doi: 10.1145/2816795.2818013
- [13] N. Mahmood, N. Ghorbani, N. F. Troje, G. Pons-Moll, and M. J. Black. AMASS: Archive of motion capture as surface shapes. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 5441–5450, Oct. 2019. doi: 10.1109/ICCV.2019.00554
- [14] M. Parger, J. H. Mueller, D. Schmalstieg, and M. Steinberger. Human upper-body inverse kinematics for increased embodiment in consumer-grade virtual reality. In *Proceedings of the 24th ACM Symposium on Virtual Reality Software and Technology, VRST '18*. Association for Computing Machinery, New York, NY, USA, 2018. doi: 10.1145/3281505.3281529
- [15] T. von Marcard, B. Rosenhahn, M. J. Black, and G. Pons-Moll. Sparse inertial poser: Automatic 3d human pose estimation from sparse imus. *CoRR*, abs/1703.08014, 2017.
- [16] A. Winkler, J. Won, and Y. Ye. Questsim: Human motion tracking from sparse sensors with simulated avatars. In *SIGGRAPH Asia 2022 Conference Papers*, pp. 1–8, 2022.
- [17] D. Yang, D. Kim, and S.-H. Lee. LoBSTR: Real-time Lower-body Pose Prediction from Sparse Upper-body Tracking Signals. *Computer Graphics Forum*, 2021. doi: 10.1111/cgf.142631
- [18] Y. Ye, L. Liu, L. Hu, and S. Xia. Neural3points: Learning to generate physically realistic full-body motion for virtual reality users. *arXiv preprint arXiv:2209.05753*, 2022.
- [19] X. Yi, Y. Zhou, M. Habermann, S. Shimada, V. Golyanik, C. Theobalt, and F. Xu. Physical inertial poser (pip): Physics-aware real-time human motion tracking from sparse inertial sensors. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2022.
- [20] X. Yi, Y. Zhou, and F. Xu. Transpose: Real-time 3d human translation and pose estimation with six inertial sensors. *ACM Trans. Graph.*, 40(4), jul 2021. doi: 10.1145/3450626.3459786